



2013 Assessment Examination

FORM VI

MATHEMATICS EXTENSION 1

Monday 20th May 2013

General Instructions

- Writing time — 1 hour 30 minutes
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 55 Marks

- All questions may be attempted.

Section I – 7 Marks

- Questions 1–7 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II – 48 Marks

- Questions 8–11 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Eight.
- Write your candidate number on this question paper and submit it with your answers.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Candidature — 113 boys

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SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

When $2x^3 + x^2 + kx - 4$ is divided by $(x - 1)$ the remainder is 2. The value of k is:

1

- (A) -7
- (B) 7
- (C) 1
- (D) 3

QUESTION TWO

The velocity v of a certain object is related to the displacement x by $v^2 = (x - 3)^2$. Its acceleration is:

1

- (A) $x - 3$
- (B) $2(x - 3)$
- (C) $\frac{1}{6}(x - 3)^3$
- (D) $\frac{1}{3}(x - 3)^3$

QUESTION THREE

A particle starts at rest 3 m to the right of the origin and moves in simple harmonic motion along the x -axis with period 2 s. The equation of motion is:

1

- (A) $x = 3 \sin 2t$
- (B) $x = 3 \cos 2t$
- (C) $x = 3 \sin \pi t$
- (D) $x = 3 \cos \pi t$

QUESTION FOUR

The equation $x^3 - 2x^2 - x + 1 = 0$ has roots α , β and γ . Which of the following is true?

1

- (A) $\alpha + \beta + \gamma = -2$ and $\alpha\beta\gamma = -1$
- (B) $\alpha + \beta + \gamma = -2$ and $\alpha\beta\gamma = 1$
- (C) $\alpha + \beta + \gamma = 2$ and $\alpha\beta\gamma = -1$
- (D) $\alpha + \beta + \gamma = 2$ and $\alpha\beta\gamma = 1$

Exam continues next page ...

QUESTION FIVE

The exact value of $\tan \left(\cos^{-1} \left(-\frac{2}{3} \right) \right)$ is:

1

- (A) $\frac{\sqrt{5}}{2}$
- (B) $-\frac{\sqrt{5}}{2}$
- (C) $\frac{2}{\sqrt{5}}$
- (D) $-\frac{2}{\sqrt{5}}$

QUESTION SIX

The double angle formulae can be used to show that $x = 3 - 4\sin^2 t$ is an example of simple harmonic motion. The centre of motion is:

1

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 5$

QUESTION SEVEN

A steel ball is fired at an initial speed of 12 m/s at an angle of elevation of 30° under the influence of gravity alone. The speed of the ball at its maximum height is:

1

- (A) 0 m/s
- (B) 6 m/s
- (C) $6\sqrt{3}$ m/s
- (D) 12 m/s

————— End of Section I —————

Exam continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

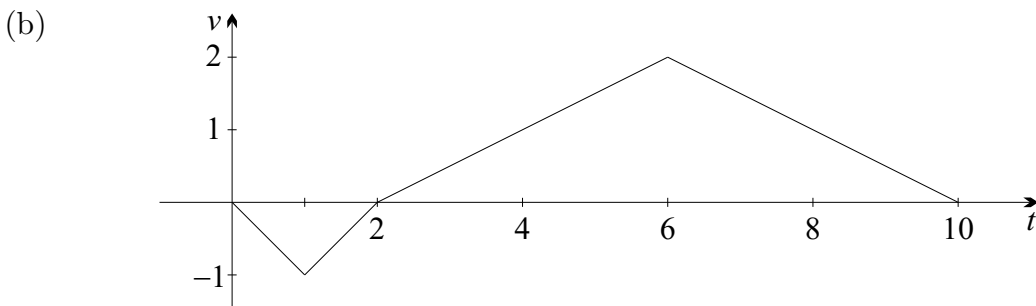
QUESTION EIGHT (12 marks) Use a separate writing booklet. Marks

- (a) Use the factor theorem to show that $(x + 1)$ is a factor of $P(x) = 6x^3 + 5x^2 - 2x - 1$. 1
- (b) Let $f(x) = 4\sin^{-1} 2x$. What is the domain of $f(x)$? 1
- (c) Differentiate $y = \tan^{-1}(3x)$. 2
- (d) The equation $x^3 + 2x^2 - 1 = 0$ has roots α , β and γ .
- (i) Write down the values of $\alpha\beta\gamma$ and $\alpha\beta + \alpha\gamma + \beta\gamma$. 1
- (ii) Evaluate $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. 1
- (e) Evaluate $\int_0^{\sqrt{3}} x\sqrt{x^2 + 1} dx$ by using the substitution $u = x^2 + 1$. 3
- (f) Let $\frac{3x^3 - x^2 + 2x + 1}{x^2 - 1} = P(x) + \frac{R(x)}{x^2 - 1}$. 3

Determine the polynomials $P(x)$ and $R(x)$ by long division.

QUESTION NINE (12 marks) Use a separate writing booklet. Marks

- (a) The polynomial $P(x)$ has degree 3 and the zeroes are -1 , 1 and 2 . The graph of $y = P(x)$ passes through the point $(3, 16)$. Find $P(x)$ in factored form. 2



The graph above shows the velocity–time graph of an object initially at the origin.

- (i) When is the object stationary? 1
- (ii) During what period is the acceleration positive? 1
- (iii) At what time does the object return to the origin? 1
- (iv) At what time is it furthest from the origin? 1

Exam continues next page ...

QUESTION NINE (Continued)

- (c) A tennis player serves a ball from a height of 1.8 m. The ball initially travels horizontally with a velocity of 35 m/s. Neglect air resistance and assume that the acceleration due to gravity is 10 m/s^2 . Let x metres be the horizontal distance the ball has travelled and let y metres be its height at times t seconds.

(i) Show by integration that the equations of motion of the ball are

2

$$x = 35t \quad \text{and} \quad y = 1.8 - 5t^2.$$

(ii) Find how long it takes for the ball to hit the ground.

1

(iii) What is the horizontal distance travelled in that time?

1

(iv) By how much does the ball clear the net which is 0.95 m high and 14 m from the player? (Ignore the dimensions of the ball.)

2

QUESTION TEN (12 marks) Use a separate writing booklet.

Marks

- (a) The polynomial $P(x)$ is monic and has degree 3. It has a quadratic factor $(x^2 - 1)$. When $P(x)$ is divided by $(x - 2)$ the remainder is -9 .

3

Find $P(x)$ and hence solve the equation $P(x) = 0$.

- (b) A car accelerates away from the origin so that its position x metres at time t seconds is given by

$$x = 20(t + e^{-0.25t}).$$

(i) Find the velocity v and acceleration $\ddot{x}$ as functions of t .

2

(ii) What is the eventual speed of the car?

1

(iii) Sketch the velocity–time graph.

1

- (c) A particle is moving about the origin according to the rule

$$x = 4 \sin(3t + \frac{\pi}{4}),$$

where x is the displacement in centimetres from the origin at time t seconds.

(i) Show that this is simple harmonic motion by showing that $\ddot{x} = -n^2x$ for some value of n , and state the value of n .

2

(ii) Write down the amplitude and period of the motion.

1

(iii) Determine when the particle is at $x = 2$ for the first time.

2

QUESTION ELEVEN (12 marks) Use a separate writing booklet.

Marks

- (a) The roots of the equation $3x^3 - 19x^2 + 38x - 24 = 0$ form a geometric sequence. Solve the equation. 4

- (b) The equations of motion of a projectile fired at speed V and angle of elevation θ are:

$$x = Vt \cos \theta$$

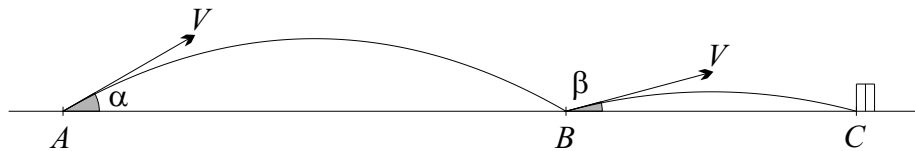
$$y = Vt \sin \theta - \frac{1}{2}gt^2,$$

where x is the horizontal distance in metres, y is the height in metres, and t is the time in seconds.

- (i) Find the time of flight. 1

- (ii) Hence show that the range is $\frac{V^2 \sin 2\theta}{g}$ metres. 1

(c)



In a game of cricket, a fielder at A on the boundary throws a cricket ball at speed V and angle of elevation α to another fielder at B . The fielder at B instantly relays the throw at the same speed, but at an angle of elevation β , to a wicket-keeper at C . The three points A , B and C are collinear. The situation is shown in the diagram above.

Use your answers to part (b) to help answer the following.

- (i) Write down an expression for the total time taken for the ball to arrive at C . 1
- (ii) Write down an expression for the total distance AC . 1
- (iii) The fielder at A throws another cricket ball at the same speed V and at an angle of elevation γ directly to the wicket-keeper at C . In the following you may assume that all three angles α , β and γ are less than 45° and greater than 0° .
- (α) Explain why $\gamma > \alpha$ and $\gamma > \beta$. 1
- (β) Show that $\sin 2\gamma = \sin 2\alpha + \sin 2\beta$. 1
- (γ) Show that it takes longer to throw the ball directly to the wicket-keeper than to relay it via B . 2

————— End of Section II —————

END OF EXAMINATION

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

CANDIDATE NUMBER:

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Multiple Choice (with possible source of errors)

Q 1 (D) (A) $P(-1) = 2$, (B) $P(-1) = 2$ sign error, (C) $k - 1 = 2$, then $k = 1$

Q 2 (A) (B) $\frac{d}{dx} v^2$, (C) $\int \frac{1}{2} v^2 dx$, (D) $\int v^2 dx$

Q 3 (D) (A) $x(0) = 0$, $n = T$, (B) $n = T$, (D) $x(0) = 0$

Q 4 (C) (A) $\frac{b}{a}$, $-\frac{d}{a}$, (B) $\frac{b}{a}$, $\frac{d}{a}$, (D) $-\frac{b}{a}$, $\frac{d}{a}$

Q 5 (B) (A) wrong quadrant, (C) reciprocal, wrong quadrant, (D) reciprocal

Q 6 (A) (B) amplitude, (C) misread: $3 - 4 \sin t$, (D) add amplitude to 3

Q 7 (C) (A) $\dot{y}$, (B) $\dot{y}(0)$, (D) initial speed

QUESTION EIGHT (12 marks)

(a) $P(-1) = -6 + 5 + 2 - 1$
 $= 0$ so $(x + 1)$ is a factor



(b) $-1 \leq 2x \leq 1$
 so $-\frac{1}{2} \leq x \leq \frac{1}{2}$



(c) $y = \tan^{-1} 3x$
 so $y' = \frac{3}{1 + 9x^2}$



[(1) for derivative of $\tan^{-1}$ and (1) for chain rule.]

(d) (i) $\alpha\beta\gamma = 1$
 $\alpha\beta + \alpha\gamma + \beta\gamma = 0$



(ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma}$
 $= 0$



(e) $I = \int_0^{\sqrt{3}} x\sqrt{x^2+1} dx$

Let $u = x^2 + 1$

at $x = 0, \quad u = 1$

$x = \sqrt{3} \quad u = 4$

and $\frac{1}{2}du = x dx,$

so $I = \int_1^4 \frac{1}{2}\sqrt{u} du$

$= \frac{1}{3} \left[u^{3/2} \right]_1^4$

$= \frac{7}{3}$



(f) $\frac{3x-1}{x^2-1} \left| \begin{array}{r} 3x^3 - x^2 + 2x + 1 \\ 3x^3 \\ \hline -x^2 + 5x + 1 \end{array} \right.$

$\frac{3x^3}{-x^2 + 5x + 1} - 3x$

$-x^2 + 5x + 1$

$\frac{-x^2}{+1}$

$5x$

so $P(x) = 3x - 1$ and $R(x) = 5x$



Total for Question 8: 12 Marks

QUESTION NINE (12 marks)

(a) Let $P(x) = a(x+1)(x-1)(x-2)$

$(3, 16)$ is on $y = P(x)$ so

$16 = a \times 4 \times 2 \times 1$

thus $a = 2$

and $P(x) = 2(x+1)(x-1)(x-2)$



(b) (i) $t = 0, 2, 10$

(ii) $1 < t < 2$ or $2 < t < 6$

[Accept $1 < t < 6$ or similar.]

(iii) By equal areas, $t = 4$

(iv) $t = 10$



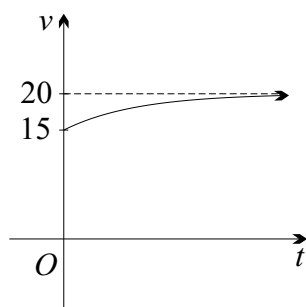
- (c) (i) $\ddot{x} = 0$ $\ddot{y} = -10$
 so $\dot{x} = C_1$ so $\dot{y} = C_3 - 10t$
 At $t = 0$, $35 = C_1$ At $t = 0$, $0 = C_3 - 0$
 so $\dot{x} = 35$ so $\dot{y} = -10t$
 thus $x = 35t + C_2$ thus $y = C_4 - 5t^2$
 At $t = 0$, $0 = 0 + C_2$ At $t = 0$, $1.8 = C_4 - 0$
 so $x = 35t$ so $y = 1.8 - 5t^2$ ☒
- (ii) At $y = 0$ $5t^2 = \frac{9}{5}$ ☒
 so $t = \frac{3}{5}$ ($t > 0$) ☒
- (iii) At $t = \frac{3}{5}$ $x = 21$ ☒
- (iv) At $x = 14$ $35t = 14$ ☒
 so $t = \frac{2}{5}$
 Now $y(\frac{2}{5}) = \frac{9}{5} - 5 \times (\frac{2}{5})^2$
 $= 1$
 Thus the ball clears the net by 0.05 m ☒

Total for Question 9: 12 Marks

QUESTION TEN (12 marks)

- (a) Let $P(x) = (x^2 - 1)(x - \alpha)$ ☒
 Now $P(2) = -9$
 so $3(2 - \alpha) = -3$
 thus $\alpha = 5$ ☒
 hence $P(x) = (x + 1)(x - 1)(x - 5)$.
 Thus $P(x) = 0$ has solutions $x = -1, 1, 5$. ☒
- (b) (i) $v = 20(1 - \frac{1}{4}e^{-0.25t})$ ☒
 $= 5(4 - e^{-0.25t}) \text{ m s}^{-1}$ ☒
 $\ddot{x} = \frac{5}{4}e^{-0.25t} \text{ m s}^{-2}$ ☒
- (ii) $\lim_{t \rightarrow \infty} v = \lim_{t \rightarrow \infty} 5(4 - e^{-t/4})$
 $= 5(4 - 0)$
 $= 20 \text{ m s}^{-1}$ ☒

(iii)



(c) (i) $x = 4 \sin(3t + \frac{\pi}{4})$
 $\dot{x} = 12 \cos(3t + \frac{\pi}{4})$
 $\ddot{x} = -36 \sin(3t + \frac{\pi}{4})$
 $= -9x$



with $n = 3$



(ii) amplitude = 4

period = $\frac{2\pi}{3}$



(iii) At $x = 2$, $\sin(3t + \frac{\pi}{4}) = \frac{1}{2}$

so $3t + \frac{\pi}{4} = \frac{\pi}{6}, \frac{5\pi}{6}, \dots$



thus $t = -\frac{\pi}{36}, \frac{7\pi}{36}, \dots$

Hence the first positive solution is $t = \frac{7\pi}{36}$



Total for Question 10: 12 Marks

QUESTION ELEVEN (12 marks)

(a) Let the roots be $\frac{a}{r}$, a and ar , then:

$$\frac{a}{r} \times a \times ar = \frac{24}{3} \quad (\text{product of roots})$$

so $a = 2$



and $\frac{a}{r} + a + ar = \frac{19}{3} \quad (\text{sum of roots})$

or $6r^2 - 13r + 6 = 0$



thus $(3r - 2)(2r - 3) = 0$

so $r = \frac{2}{3}$ or $\frac{3}{2}$



In both cases the roots are $\frac{4}{3}, 2, 3$



(b) (i) At $y = 0$ $t(V \sin \theta - \frac{1}{2}gt) = 0$

so $t = \frac{2V \sin \theta}{g} \quad (t \neq 0)$



$$\begin{aligned}
 \text{(ii) From (i)} \quad x &= V \times \frac{2V \sin \theta}{g} \times \cos \theta \\
 &= \frac{V^2 \times 2 \sin \theta \cos \theta}{g} \\
 &= \frac{V^2 \sin 2\theta}{g}.
 \end{aligned}$$



$$\begin{aligned}
 \text{(c) (i) Time } AC &= \frac{2V \sin \alpha}{g} + \frac{2V \sin \beta}{g} \\
 &= \frac{2V}{g}(\sin \alpha + \sin \beta)
 \end{aligned}$$



$$\begin{aligned}
 \text{(ii) Range } AC &= \frac{V^2 \sin 2\alpha}{g} + \frac{V^2 \sin 2\beta}{g} \\
 &= \frac{V^2}{g}(\sin 2\alpha + \sin 2\beta)
 \end{aligned}$$



(iii) (α) Since range $AC > \text{range } AB$

$$\sin 2\gamma > \sin 2\alpha$$



so $\gamma > \alpha$ (since 2γ and 2α are acute)

and likewise, $\gamma > \beta$.

[Accept any other valid argument.]

(β) From the range of AC

$$\frac{V^2}{g} \sin 2\gamma = \frac{V^2}{g}(\sin 2\alpha + \sin 2\beta)$$



$$\text{or } \sin 2\gamma = \sin 2\alpha + \sin 2\beta$$

$$(\gamma) \quad \sin 2\gamma = \sin 2\alpha + \sin 2\beta$$

$$\text{so } 2 \sin \gamma \cos \gamma = 2 \sin \alpha \cos \alpha + 2 \sin \beta \cos \beta$$



Now $\gamma > \alpha$ so $\cos \gamma < \cos \alpha$ (cos is decreasing for acute angles)

and $\gamma > \beta$ so $\cos \gamma < \cos \beta$ (again, cos is decreasing for acute angles)

$$\text{thus } 2 \sin \gamma \cos \gamma > 2 \sin \alpha \cos \gamma + 2 \sin \beta \cos \gamma$$

$$\text{hence } \sin \gamma > \sin \alpha + \sin \beta$$



$$\text{thus } \frac{2V}{g} \sin \alpha > \frac{2V}{g}(\sin \alpha + \sin \beta)$$

That is, the direct throw takes longer than the relayed throw.

Total for Question 11: 12 Marks

Out of interest, here is another solution to the last part.

Let $AB = a$ and $BC = b$ then the time for the relayed throw is

$$\begin{aligned} t_1 &= \frac{a}{V \cos \alpha} + \frac{b}{V \cos \beta} \\ &= \frac{a \cos \beta + b \cos \alpha}{V \cos \alpha \cos \beta} \end{aligned}$$

and the time for the direct throw is

$$t_2 = \frac{a + b}{V \cos \gamma}.$$

$$\begin{aligned} \text{Now } t_2 - t_1 &= \frac{a + b}{V \cos \gamma} - \frac{a \cos \beta + b \cos \alpha}{V \cos \alpha \cos \beta} \\ &= \frac{(a + b) \cos \alpha \cos \beta - a \cos \beta \cos \gamma - b \cos \alpha \cos \gamma}{V \cos \alpha \cos \beta \cos \gamma} \\ &= \frac{a \cos \beta (\cos \alpha - \cos \gamma) + b \cos \alpha (\cos \beta - \cos \gamma)}{V \cos \alpha \cos \beta \cos \gamma}. \end{aligned}$$

But $\gamma > \alpha$ and $\gamma > \beta$

so $\cos \alpha - \cos \gamma > 0$ and $\cos \beta - \cos \gamma > 0$

Hence $t_2 - t_1 > 0$,

that is, the direct throw is slower than the relayed throw.

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